Homework 4

William Flanders*
Haverford College
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1. NEWMAN EXERCISE 5.4

We want to explore light diffraction in telescopes. In fields like astronomy, we observe the light that is emitted from objects to study them. These objects emit light as point sources where the light then travels to our telescopes, passes through an aperture, and is focused onto the focal plane. However, interestingly, when the light passes through the aperture, instead of seeing a single dot, we see a diffraction pattern; a disc in the center with a series of rings emanating outwards. This disc is called an Airy disc pattern and its intensity can be modeled with the relationship

$$I(r) = (\frac{J_1(kr)}{kr})^2 \tag{1}$$

where r is the distance in the focal plane from the center of the diffraction pattern, $k = \frac{2\pi}{\lambda}$ and $J_1(x)$ is a Bessel function. The Bessel functions $J_m(x)$ are given by

$$J_m(x) = \frac{1}{\pi}\int_0^{\pi} cos(m\theta - xsin(\theta)d\theta \tag{2}$$

where m is a nonnegative integer and $x \geq 0$.

We will want to create a figure of this diffraction pattern in python. First, we will need to tackle the Bessel function. We can numerically solve for values of the Bessel functions using Simpson's rule for integration.

Simpson's rule can be described by

$$\int_a^b f(x)dx = \frac{h}{3}[f(x_0)+4\sum_{i=1}^{n/2} f(x_{2i-1})+2\sum_{i=1}^{n/2-1} f(x_{2i})+f(x_n)] \tag{3}$$

where $h = \frac{(b-a)}{n}$ and n is the number of steps.

We can start by writing a program to calculate the mth Bessel function, and then use this to analytically solve Equation 1. Beginning with pseudo code

```
from math import *
import matplotlib.pyplot as plt

def f(x):
    our target function
```

```
def J(m, x, theta):
    # Using Simpson's rule
    define h
    define n
    initialize sum

    for i in range (1, n):
        initialize theta
            if i is odd:
                sum += 4 * f(m, x, theta)
            if i is even:
                sum += 2 * f(m, x, theta)
    sum = sum*(h/3*pi)
    return sum

plt.plot(J_0, J_1, and J_2 against x)
```

After running our program, Figure 1 shows our calculation for the Bessel functions $J_0, J_1, and J_2$ plotted against x. We will be using J_1 in our calculation for Equation 1.

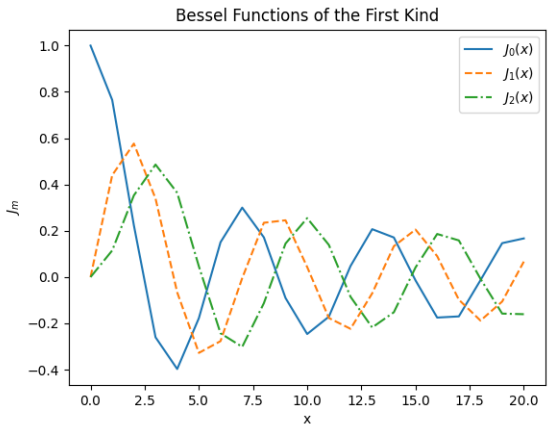

FIG. 1: Bessel functions $J_m(x)$ of the first kind for $m = 0, 2, 3$ plotted against x.

For our program that calculates the intensity and creates a figure, we will want to utilize our program that finds J_1 and then create a density plot using the matplotlib library. Beginning with pseudo code

```
from math import *
import matplotlib.pyplot as plt
import numpy as np

J(m, x, theta):
    Bessel function calculation using Simpson's r
```

*Electronic address: wflanders@haverford.edu; URL:

```
I(r):
    # Function that calculates Equation 1
    return I

grid = an empty N x N grid
separation = space between x and y values
loop i across x values:
    loop j across y values:
        calculate r

        if r < 10**-8:
            grid[i, j] = 1/2
        else:
            grid[i, j] = I(r)
plot(grid as a heatmap)
```

Importantly, we need to consider when $r = 0$ for Equation **??**. Mathematically, this value for the intensity is undefined as we would be dividing by zero. This would give us a divide by zero error in our program anywhere in the grid where r is 0. However, $\lim_{x\to 0} \frac{J_m(x)}{x} = \frac{1}{2}$. This solves our divide by 0 issue, and we can incorporate the limit via way of an if statement. The resulting figure for the diffraction pattern is shown in 2.

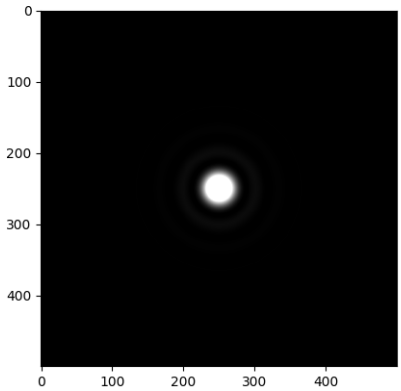

FIG. 2: Diffraction pattern for light through an aperture.

The disc shape is clearly visible with rings that radiate outwards and become more faint the further from the center we go.

2. NEWMAN EXERCISE 5.9

We want to investigate how heat capacity varies with respect to temperature in solids. The relationship can be modeled from Debye's theory of solids where the heat capacity, C_V is given by

$$C_V = 9V\rho k_B(\frac{T}{\theta_D})^3 \int_0^{\theta_D/T} \frac{x^4 e^x}{(e^x-1)^2} dx \tag{4}$$

where V is the volume of the solid, ρ is the number density of atoms, k_B is Boltzmann's constant, T is the temperature, and θ_D is Debye temperature. The Debye temperature is a property of solids that depends on their density and speed of sound. We will want to use Equation 3 to model the heat capacity of 1000 cubic centimeters of Aluminum at various temperatures. To handle the integral, we will use Gaussian quadrature.

Gaussian quadrature is a method of computation that relies on a weighted sum of values of the function at specific points to output a reliable value of the integral. We can utilize Newman's gaussxwab method to handle the computation of the weights and the specific x values for our sum.

We start with pseudo code

```
import matplotlib.pyplot as plt
import numpy as np
from gaussxw import gaussxwab

def f(x):
    our integrand function for integration

def cv(T):
    # Function that calculates the heat capacity
    Initialize variables
    wp, xp = gaussxwab(n, lower limit, upper limi

    sum = 0
    loop through n calculating the weighted sum
    calculate final value

plt.plot(t, cv)
```

Our Gaussian quadrature computation will use 50 sample points and have boundaries as defined by Equation 3. The values of our constants: $V = 1000cm^3$, $\rho = 6.022 \cdot 10^{28} m^{-3}$, $\theta_D = 428K$ and we will be looking at temperatures from $5K$ to $500K$. The resulting figure is showing in Figure 3.

We can see that heat capacity does vary with respect to temperature. As temperature of the Aluminum increases, holding all else constant, the energy required to increase the Aluminum's temperature also increases. There is a steep increase from around $50K$ t0 $300K$ before the curve flattens out.

3. SURVEY

The problem set took roughly 5-6 hours. Exercise 5.9, with the help of Newman's Gaussian quadrature, was relatively simple to solve, both in terms of computation

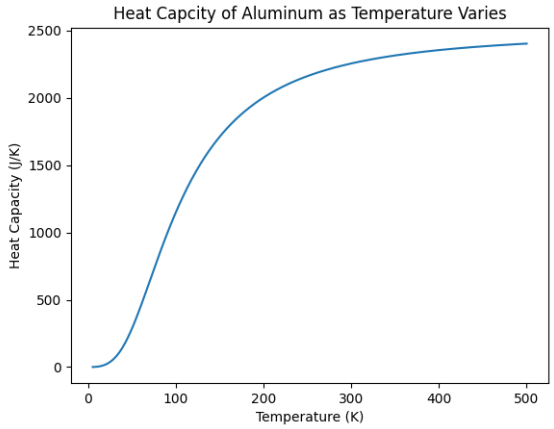

FIG. 3: Heat capacity of Aluminum as temperature varies from 5K to 500K.

and the problem's demands. Exercise 5.4 was somewhat challenging mostly due to creating the density plot and the relatively long time for my computer to finish running the program.